

STANDARD OPERATING PROCEDURE

PRODUCTION ISSUE : ROOT CAUSE FIXTURE

REVISION HISTORY

Revision No./Version No.	Date	Prepared by	Approved by	Details of Change
1.0	01-July-2025	Tanvi Samant (Software Developer)	Prachi Naik (Assistant Manager IT)	Initial Release

OBJECTIVE

To define a structured approach for identifying the root cause of software issues and ensuring their permanent resolution to prevent recurrence, thereby maintaining product quality and client satisfaction.

PARTICIPANTS

Technical Lead(TL)/Maintenance Lead(ML): Oversees root cause identification and reviews fixtures.

Developer/Maintenance Engineer: Investigates issues and implements fixes.

Implementation Engineer: Explains the scenarios faced on live and checks the same after fixture.

Testing Team: Will Test the changes done.

PROCEDURE

Ensure every production issue or backend update ticket is thoroughly analyzed to determine whether a permanent fix is required.If so, follow below procedure:

1. Identify Root Cause

1. Create a "Production Issue Root Cause Fixture" in GELJIRA & link it with Production Issue.
2. Thoroughly identify where, why and when the issue is happening.
3. Replicate the issues in staging or test environment to confirm the issue and find out how much data might have same issue in past.
4. Check how the issue can be fixed permanently.
5. Discuss with TL/ML how the issue can be fixed and how much efforts will be needed to fix.
6. TL to consult with PM of efforts to fix issues are more that 24 hours.
7. If confirmed by TL/ML proceed with fixture.

2. Fix Design and Test

1. Develop the fix.
2. Ensure the fix addresses both current and past cases.
Example: If the issue is related to posting failures on Goa Online due to a null Service ID being passed, the solution should ensure that Service ID is never passed as null going forward. Additionally, all past failed postings should be corrected through backend updates.
3. The fix does not introduce any new issues.
4. Adheres to all coding and database standards.
5. Perform unit testing and validate the solution against various test cases.
6. Develop plan to update the existing data or intimate user etc
7. Ensure the fix is tested by testing team on Test environment.
8. Ensure that all code and database changes are reviewed by the respective Tech Lead and/or Database Vertical, as per the established review procedures.
9. Maintain detailed documentation of the analysis, solution design, and testing performed.
Attach this documentation to the "Production Issue Root Cause Fixture" ticket in GEL JIRA.
10. After successful testing and review clearance, deploy the changes to the Production environment following the standard deployment policy.

3. Post-Deployment Monitoring

- After deployment, monitor the production environment.
- Ensure that the issue does not reoccur.
- Ensure to close the ticket once the issue is fixed and tested.

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Logging in Jira/TM

Issue Type*: Production Issue Root Cause Fixture

Assignee*: TL/Developer

Reporter (ticket to be created by) *: TL/ML

Priority*: High/Medium/Low

Description: Fill in the details requested as per below

1. What issue noticed during production issue:
2. What is the possible reason towards the issue:

Link To: Production Issue/Backend Data Update Request or Backend Report Request ticket if if available.

Work Logging by*: TL/ML/Developer

Note: A Production Issue Root Cause Fixture ticket should be closed by the Reporter only after verifying that a permanent fix has been implemented for the issue.

Workflow for Production Issue Root Cause Fixture

